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NCAA-D-26-02775R2 - Editor Decision

1 message

Neural Computing and Applications (NCAA) <em@editorialmanager.com>

Wed, Aug 26, 2026 at
7:01 PM

Reply-To: "Neural Computing and Applications (NCAA)" <mangayarkarasi.lakshmiram@springer.com>

To: Prakash Ukhalkar <prakashu.mca@gmail.com>

CC: "Ravikant Zirmite" rsz.imcc@mespune.in, "Manasi Bhate" msb.imcc@mespune.in

Dear Mr Ukhalkar,

We have received the reports from our advisors on your manuscript, "India VIX Regime-Conditional Directional Prediction of Bank Nifty: A Stacking Ensemble Approach", which you submitted to Neural Computing and Applications.

Based on the advice received, the Editor feels that your manuscript could be reconsidered for publication should you be prepared to incorporate the major revisions recommended by the reviewers.

When preparing your revised manuscript, you are asked to carefully consider the reviewer comments which are attached, and submit a list of responses to the comments.

Your list of responses should be uploaded as a file in addition to your revised manuscript.

Your revised manuscript will be sent back to the reviewers to allow them to confirm that the changes address the issues they have set out in their review comments, and that the revised manuscript is acceptable for publication.

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Your username is: prakashu.research

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Please click "Author Login" to submit your revision.

If you need any further help, please use the "Contact Us" link in the banner at the top of the Editorial Manager website.

We look forward to receiving your revised manuscript before 23 Oct 2026.

Best regards,
Ajay Kaul, Ph.D
Associate Editor
Neural Computing and Applications

COMMENTS FOR THE AUTHOR:

Reviewer #2: I have gone through the comments in the previous review and the response. The authors have put significant effort in revising and improving the article. Therefore, I have no further comments and recommend it be accepted in its current form.

Reviewer #4: The paper tests whether India VIX conditions 21-day directional predictability in Bank Nifty, using a stacking ensemble of four tree learners and a BiLSTM over sixteen indicators. Since the last round the authors have fixed the split boundary, made the regime threshold pre-test, and moved to dependence-aware inference.

A lot of work went into this revision and I want to say so first. Much of the body was rewritten and most tables carry new numbers. The models were genuinely retrained: the previous version reported 21-day accuracy of 0.6101 on 277 rows, which is 169 correct, and this one reports 0.6758 on 256, which is 173, and you cannot raise the number of correct calls by dropping rows. The authors then published numbers much worse for their own hypothesis than the ones they started with, and said so plainly in the abstract, Section 5.3 and the Conclusion. That is uncommon and it deserves recognition.

My problem is with what the corrections leave. Under the pre-test threshold the High-VIX subset has eight observations and all eight are Up. The local majority baseline is therefore 100 per cent, and the model's 100 per cent accuracy is exactly what a constant Up rule would score. Low-VIX accuracy is 66.5 per cent against a 70.2 per cent baseline, and overall accuracy is 67.6 per cent against 71.1 per cent. The regime difference is not significant under the non-overlapping Fisher test or the block permutation test, and the trading rule executes a single trade.

On eight single-class observations the design cannot test the hypothesis either way. I am not saying it is disproved. I am saying this window can no longer speak to it, and rewriting will not change that. Keeping a positive claim would need an evaluation window with several stress episodes, which is new work.

There is also a practical problem. The corrected analysis went into the text and most tables, but into no figure and not into the older rows of several tables, so the paper now reports two incompatible sets of results. The points below mostly follow from that.

1. The framing still argues the old case, sometimes with the old numbers. The Introduction says "the answer is yes, but only at the 21-trading-day horizon", and that the study "shows that selective participation improves risk-adjusted performance relative to buy-and-hold while using market exposure on only 48 out of 277 test days". Table 7 gives $p = 1.000$ at that horizon and Table 8 gives one trade over 21 of 256 days, behind buy-and-hold. Section 5.2 calls the regime filter "the core expert-system contribution of the paper" between two passages that state the baseline problem correctly. Sections 2.5, 4 and 7 read the same way, Table 1 still specifies the partition by "contemporaneous India VIX", and Section 3.4 keeps a paragraph defending the threshold it has just replaced.

2. No figure in Sections 5 or 6 appears to have been rebuilt. Figure 3 plots 0.845 against 0.548 with two asterisks for $p < 0.004$, where Table 7 a few lines above gives 1.000 against 0.665 at $p = 1.000$. Figure 5 labels its threshold line "VIX p75 threshold (15.08)" and shades High-VIX episodes repeatedly from January to June 2025 with the strategy trading through them, where Section 5.3 describes one 21-day trade. Figure 7 runs from a Sharpe of about 2.18 to about 1.90 against a buy-and-hold line at 1.89, so it puts the strategy ahead of the benchmark at every cost level, where Table 9 puts it behind at every level. Figure 8 is labelled $n = 58$ and $n = 219$, and Figure 9 carries "HV obs:

84.5%". Figure 4 gives a 21-day High-VIX AUC of 0.6440, which Section 5.2 calls "the only material threshold-free ranking advantage"; if all eight outcomes are Up then that subset has one class and AUC is undefined.

3. Table 10 gives the VIX-High and GARCH-High intersection as 47 where VIX-High has 8 members, and at 87.2 per cent where the parent set is 100 per cent. Its VIX rows total 256 and its GARCH rows 277. In Table 11, p75 selects 8 days and p85 selects 28, which no single distribution allows, and the 2025-Q1 and Q2 rows give 17 and 41 High-VIX days against a stated total of 8. Those two sum to 58, the superseded count.

4. The shorter horizons look like pre-correction output. Table 6's 1-day and 5-day rows are unchanged from the previous version to four decimal places, and Table 7's entry for entry. Table 7's rows total 276, 272 and 256 observations, with High-VIX counts of 58, 57 and 8. After a split change of about 21 rows and a full retrain I cannot see how those could be unchanged. This matters because the claim that no effect appears at the shorter horizons is comparing a corrected partition against two uncorrected ones.

5. Several analyses named in the response letter are not in the file I received. SHAP, the ablation study cited to a Section 4.4 that does not exist, the year-wise walk-forward, per-regime precision, recall and F1, the tuned hyperparameters and the cross-validation protocol, the Sharpe conventions, the per-learner baseline comparison, and the feature formulas. Neither "SHAP" nor "ablation" occurs in the text. If these went in as separate supplementary files that did not reach me I withdraw the point. Otherwise I assume the work was done and not merged, since the class metrics quoted in the letter reconcile exactly with the manuscript's own accuracy figures. The walk-forward is the one I most want to see. The letter reports the High-VIX advantage as positive in 2021 and negative in 2019, 2020, 2022, 2024 and 2025, and no reader can get to it.

6. The Sharpe conventions are not stated. Section 5.3 gives the execution timing but not the risk-free rate or the annualisation basis. The corrected conclusion that the strategy trails buy-and-hold rests entirely on 0.455 against 1.204, so both series need to be shown to be computed the same way. Table 9's annualised column also looks like it uses a divisor of 276 rather than the 256-row sample. In any case a Sharpe ratio built from one position across 256 days, 235 of them flat, is not a distribution statistic, and reporting the trade itself would tell the reader more.

7. Section 6.2 concludes that "the data indicate that India VIX is the stronger gatekeeper". In Table 10 the GARCH split is the only one with a nominally significant effect, at $p = 0.039$, against $p = 1.000$ for VIX.

8. Section 6.1 says a system that only looked good before costs would not be credible, "but that is not the pattern observed here", in a passage showing the strategy behind buy-and-hold at every cost level. The same paragraph puts the constant drawdown down to the filter reducing exposure, where it follows from there being one position. Section 6.3 reports a bootstrap interval excluding zero alongside tests that are not significant. It does not hide the tension but does not explain it, and the permutation p-value is not printed.

9. Smaller things. Section 5.4 reads greater confidence into a 0.023 gap in mean probability across eight observations. The naive-majority rate is 71.1 per cent in Table 6 and 68.9 per cent in Figures 8 and 9. Section 6.3 gives the old threshold as yielding 56 High-VIX days where the letter and Table 7 use 58. The letter moves the training boundary a month earlier while the row count rises from 1,410 to 1,794, which needs a sentence; I suspect the current figure is right. The four graph-neural-network references were added at a previous reviewer's request, so this is no criticism of the authors, but none concerns financial time series and all four hang on one sentence. The block bootstrap and permutation procedures are not cited to any source.

I would welcome access to the analysis code, in particular the split construction, the threshold calculation and the trading simulation.

One last thought. Moving a single threshold from a test-derived value to a pre-test one took the High-VIX sample from 58 days to 8 and removed the significance. Fixing the split and the execution protocol took the strategy Sharpe from 2.216 to 0.455 while the benchmark fell from 1.963 to 1.204, reversing the comparison. Two ordinary mistakes accounted for the whole of the original result, and the authors are in a rare position to show exactly how much, on their own claim, with the intermediate states already computed. A paper built around that, with the walk-forward at its centre and a window covering more than one stress episode, would be worth reading, and I think it is what this work should become. It would need a different title. I would be glad to see it.

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